Will Gregory

william-gregory.github.io
william.gregory@ucl.ac.uk
github.com/William-gregory
linkedin.com/in/will-gregory
scholar.google.com

Education

2017-2021	PhD in Polar Climate Science and ML, *Pass (no corrections)*	University College London (UCL)
2013-2014	MSc in Petroleum Geophysics, *Distinction*	Imperial College London
2010-2013	BSc in Geology with Geophysics, *1st class*	University of Leicester

Employment

06/2026-present — **Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellow | UCL**

Project: "Data Assimilation and Machine learning for online Parameter Estimation (DAMPEn)", will utilise large-scale remote sensing observations to learn state-dependent sea ice model physics parameters. In collaboration with the UK Met Office and the University of Tromsø.

01/2022-05/2026 — **Postdoctoral Researcher | Princeton University & Geophysical Fluid Dynamics Lab**

Funded by Multi-scale Machine Learning In coupled Earth System modeling (M^2LInES) initiative

Past Responsibilities:

- Lead coupled ocean-sea-ice emulation development through M^2LInES–AI2 partnership
- Lead hybrid sea ice modelling development in M^2LInES and GFDL
- Lead organiser of the 2025 M^2LInES annual meeting
- Co-organiser of biweekly project meetings

11/2014-07/2017 — **Depth-imaging Geophysicist | Petroleum Geo-Services Ltd**

Client-facing role in the private sector, delivering geophysical services to oil and gas companies

Past Responsibilities:

- Develop sub-surface seismic velocity models using tomographic inversion techniques
- Deliver project presentations and lead data visualisation sessions with clients
- Develop and maintain Gantt charts, status reports, and end-of-project reports.
- Train new staff in sub-surface imaging theory and in-house software

Research Funding

NERC AI for Environmental Science: Phase 1 (2025)	Researcher Co-I; led the conceptualisation and writing of the proposal vision and approach, centred on developing a next-generation coupled ocean-sea-ice emulator for investigating climate tipping points. *Pending*
Marie Curie Postdoctoral Fellowship (2025)	PI; designed and authored a full independent research programme on AI for climate model parameterisation, including scientific objectives, methodology, personal and professional development, and outreach and dissemination. *Accepted* | *98%*

Selected Honours

Fellowships

2026-2028	Marie Skłodowska-Curie Actions (MSCA) fellowship | Awarded €260,347 for 2-year postdoctoral fellowship
2017-2021	London NERC Doctoral Training Partnership (DTP) | Funding for 4-yr PhD, amounting to over £91,000

Awards

2025	Wiley top viewed article award | Gregory et al in top 10% of most-viewed 2023 papers in *JAMES*
2015	Chancellor's Masters Scholarship, University of Sussex | £3,000 to study MSc Cosmology [declined]
2013	British Petroleum Scholarship, Imperial College London | £25,000 to study MSc Petroleum Geophysics
2013	Shell Geophysics Prize | Awarded for highest contributions to BSc Geophysics programme

Scientific Contributions

Pre-prints

M Bushuk, D Bonan, S Griffies, **W Gregory**, Y Zhang, B Hurlin, Y Chen, T Rackow, HF Goessling, "Historical and projected Antarctic sea ice trends across high-resolution coupled model hierarchies," accepted in *Geophysical Research Letters*, (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, SK Clark, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna, "FloeNet: A mass-conserving global sea ice emulator that generalizes across climates," arXiv [accepted in *Geophysical Research Letters*], (2026)

L Zanna, **W, Gregory**, P Perezhogin, A Sane, C Zhang, A Adcroft, M Bushuk, C Fernandez-Granda, B Reichl, D Balwada, J Busecke,W Chapman, A Connolly, D Du, K Everard, F Falasca, R Falga, D Kamm, E Meunier, Q Liu, A Nasser, M Pudig, A Shao, JL Simpson, L Vogt, J Wu, "A framework for hybrid physics-AI coupled ocean models," arXiv, (2026)

Y Zhang, M Bushuk, M Winton, **W Gregory**, B Hurlin, L Jia, F Lu, "Subseasonal forecast improvements from sea ice concentration data assimilation in the Antarctic," EGUsphere [in review at *The Cryosphere*], (2025)

Peer-reviewed

J Duncan, E Wu, S Dheeshjith, A Subel, T Arcomano, SK Clark, B Henn, A Kwa, J McGibbon, A Perkins, **W Gregory**, C Fernandez-Granda, J Busecke, O Watt-Meyer, B Hurlin, A Adcroft, L Zanna, C Bretherton, "SamudrACE: Fast and accurate coupled climate modeling with 3D ocean and atmosphere emulators," *Geophysical Research Letters*, 53, e2025GL119340, (2026)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna, C McHugh, L Jia "Advancing global sea ice prediction capabilities using a fully coupled climate model with integrated machine learning," *Science Advances* (12), 1, eady8957, (2026)

D Balwada, R Abernathey, S Acarya, A Adcroft, J Berner, V Balaji, M Bhouri, J Bruna, M Bushuk, W Chapman, A Connolly, J Deshayes, C Fernandez-Granda, P Gentine, A Gorbunova, **W Gregory**, A Guillaumin, S Gupta, M Holland, JE Johnson, J Le Sommer, Z Li, N Loose, F Lu, P O'Gorman, P Perezhogin, B Reichl, A Ross, A Sane, S Shamekh, T Verma, J Yuval, L Zampieri, C Zhang, L Zanna "Learning machine learning with Lorenz-96," *Journal of Open Source Education* 7, 241, (2024)

W Gregory, R MacEachern, S Takao, IR Lawrence, C Nab, M Deisenroth, M Tsamados "Scalable interpolation of satellite altimetry data with probabilistic machine learning," *Nature Communications*, 15, 7453, (2024)

M Bushuk, S Ali, D Bailey, Q Bao, L Batté, U Bhatt, E Blanchard-Wrigglesworth, E Blockley, G Cawley, J Chi, F Counillon, PG Couolmbe, RI Cullather, FX Diebold, A Dirkson, E Exarchou, M Göbel, **W Gregory**, V Guemas, L Hamilton, B He, S Horvath, M Ionita, JE Kay, E Kim, N Kimura, D Kondrashov, Z Labe, W Lee, Y Lee, C Li, X Li, Y Lin, Y Liu, W Maslowski, F Massonnet, WN Meier, WJ Merryfield, H Myint, JC Acosta Navarro, A Petty, F Quao, D Schröder, A Schweiger, Q Shu, M Sigmond, M Steele, J Stroeve, N Sun, S Tietsche, M Tsamados, K Wang, J Wang, W Wang, Y Wang, Y Wang, J Williams, Q Yang, X Yuan, J Zhang, Y Zhang "Predicting September Arctic sea ice: a multi-model seasonal skill comparison," *Bulletin of the American Meteorological Society*, 105, E1170-E1203, (2024)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna "Machine learning for online sea ice bias correction in global ice-ocean simulations," *Geophysical Research Letters*, 51, e2023GL106776, (2024)

Y Zhang, M Bushuk, M Winton, B Hurlin, **W Gregory**, JC Landy, L Jia "Improvements in September Arctic sea ice predictions via assimilation of summer CryoSat-2 sea ice thickness observations.," *Geophysical Research Letters*, 50, e2023GL105672, (2023)

W Gregory, M Bushuk, A Adcroft, Y Zhang, L Zanna Deep learning of systematic sea ice model errors from data assimilation increments," *Journal of Advances in Modeling Earth Systems*, 15, e2023MS003757, (2023)

C Nab, R Mallett, **W Gregory**, JC Landy, IR Lawrence, R Willatt, J Stroeve, M Tsamados "Synoptic variability in satellite altimeter-derived radar freeboard of Arctic sea ice.," *Geophysical Research Letters*, 50, e2022GL100696, (2023)

W Gregory, J Stroeve, M Tsamados "Network connectivity between the winter Arctic Oscillation and summer sea ice in CMIP6 models and observations," *The Cryosphere*, 16, 1653-1673, (2022)

W Gregory, IR Lawrence, M Tsamados "A Bayesian approach towards daily pan-Arctic sea ice freeboard estimates from combined CryoSat-2 and Sentinel-3 satellite observations," *The Cryosphere*, 15, 2857-287, (2021)

W Gregory, M Tsamados, J Stroeve, P Sollich "Regional September sea ice forecasting with complex networks and Gaussian processes," *Weather and Forecasting*, 35, 793-806, (2020)

Invited Talks

"A new global sea ice emulator and the path toward fully data-driven coupled climate modelling" ORCAS seminar, Virtual (2026)

"How is AI shaping sea ice modeling and prediction research?" Polar Research Board, Fall Meeting, Virtual (2025)

"Fortran-integrated machine learning in SIS2 and SPEAR", GFDL AI workshop, GFDL, Princeton, USA (2025)

"How is AI shaping Arctic climate modeling and observational research?" US Interagency Arctic Research Policy Committee (IARPC) Webinar, Virtual (2025)

"Bridging AI, remote sensing, and climate modeling to understand polar climate variability and change," Institute for Advanced Computational Science (IACS) Seminar, Stony Brook University, USA (2025)

"From component to coupled: evaluating the performance of a machine-learned sea ice bias correction scheme in fully-coupled seasonal predictions," Nansen SuperIce Webinar Series, Virtual (2024)

"Towards improving numerical sea ice predictions with data assimilation and machine learning," NOAA Arctic All Hands, Vitual (2024)

"Applications of machine learning to sea ice data assimilation," 10th US Climate modeling summit, Geophysical Fluid Dynamics Laboratory, Princeton, USA (2024)

"Towards a machine-learned sea ice model parameterization from data assimilation increments," Euro-Mediterranean Center on Climate Change (CMCC) Seminar, Bologna, Italy (2024)

"Deep learning of systematic sea ice model errors from data assimilation incremenets," Courant Institute of Mathematical Sciences Guest Seminar Series, New York City, USA (2022)

"Machine learning tools for pattern recognition in polar climate science," EGU General Assembly, Vienna, Austria (2021)

"Machine learning in climate science," Government Digital Service, London, UK (2020)

Conference talks

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "Towards a mass-conservative global sea ice emulator that generalizes across climates," EGU General Assembly, Vienna, Austria (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. "Towards a mass-conservative global sea ice emulator that generalizes across climates," Climate Informatics, Lausanne, Switzerland (2026)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna, C McHugh, L Jia. "Evaluating a new hybrid sea ice model in global fully-coupled climate predictions and projections," AGU Ocean Sciences Meeting, Glasgow, UK (2026)

W Gregory, M Bushuk, J Duncan, E Wu, A Subel, S Clark, B Hurlin, O Watt-Meyer, A Adcroft, C Bretherton, L Zanna. “FloeNet: a mass-conserving emulator of the GFDL sea ice model,” CESM Polar Working Group Meeting, Boulder, USA (2026)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. “Machine-leaned sea ice bias correction in a global fully-coupled climate model,” Gordon Research Conference: Machine Learning for Actionable Climate Science, RI (2025)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. “Machine-leaned sea ice bias correction in a global fully-coupled climate model,” CESM Polar Working Group Meeting, Boulder, CO (2025)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. “Towards improving numerical sea ice predictions with data assimilation and machine learning,” Cross-VESRI Convening, Cambridge, UK (2024)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. “Towards improving numerical sea ice predictions with data assimilation and machine learning,” EGU General Assembly, Vienna, Austria (2024)

W Gregory, M Bushuk, Y Zhang, A Adcroft, L Zanna. “Machine learning for online sea ice bias correction within global ice-ocean simulations, AGU Ocean Sciences Meeting, New Orleans, USA (2024)

R MacEachern, M Tsamados, **W Gregory**, IR Lawrence, S Takao. “Fast interpolation of satellite altimetry data with probabilistic machine learning and GPU,” EGU General Assembly, Vienna, Austria (2023)

W Gregory, M Bushuk, A Adcroft, Y Zhang, L Zanna. “Deep learning of systematic sea ice model errors from data assimilation increments,” EGU General Assembly, Vienna, Austria (2023)

W Gregory, M Bushuk, A Adcroft, Y Zhang, L Zanna. “Using deep learning to predict systematic model error from sea ice data assimilation increments in a fully-coupled climate model,” AGU Fall Meeting, Chicago, USA (2022)

Teaching & Mentoring

2020	TA in BSc Geodynamics, UCL | Class size: 15-20. Format: Tutorials
2020	TA in BSc Ocean Physics, UCL | Class size: 15. Format: Tutorials
2019	TA in BSc Principles of Climate, UCL | Class size: 10-15. Format: Tutorials
2018-2019	TA in BSc Foundations of Physical Geoscience, UCL | Class size: 10-12. Format: Lectures and Tutorials
2018-2020	TA in BSc Introduction to Matlab, UCL | Class size: 15-20. Format: Lectures and Tutorials

2022 With Marc Deisinroth, co-supervised Ronald MacEachern, a postgraduate student undertaking the MSc Machine Learning course within the Department of Computer Science at UCL. I designed the MSc dissertation project, which was titled “Sea Ice Freeboard Interpolation using Gaussian Process Regression”. Ronald achieved a thesis grade of Distinction. I then led an international collaboration to build Ronald’s work into an open-source Python library, GPSat, along with the *Nature Communications* paper Gregory, MacEachern, Takao et al 2024.

Service

2026	Guest Editor for a special edition on AI-driven ocean science in BMC Marine Science (Springer Nature)
2024 & 2025	Reviewer for NSF Arctic Natural Sciences program
2024	AGU Fall Meeting Outstanding Student Presentation Award (OSPA) judge
2024	AGU Fall Meeting convener and chair for session: “Data Driven Science: Developments in Machine Learning Subgrid-Scale Parameterizations and in Reanalyses across Earth System Modeling”
2019-present	Peer reviewer for numerous international journals, including: Science Advances, Geophysical Research Letters, The Cryosphere, Journal of Advances in Modeling Earth Systems, American Meteorological Society (AMS) Journal of Climate, Climate Dynamics, Quarterly Journal of the Royal Meteorological Society, AMS AI for Earth Systems, npj Climate and Atmospheric Science

Public Engagement & Outreach

2026	Invited to roundtable on climate modelling with the UN Secretary-General's Scientific Advisory Board
2025	Early-career panelist at the Science Philanthropy Alliance meeting, titled "The Next Generation"
2023	"AI is transforming climate forecasts for melting sea ice," *Advanced Science News Article*
2023-2024	Project development chair for the Climatematch academy outreach program | Independently managed 4 individuals who were developing student project materials
2022-2023	Curriculum content reviewer for the Climatematch Academy outreach programme | Responsible for reviewing all content relating to fundamentals of climate science
2022	Presented on EDI progress within NOAA and GFDL, Princeton University and GFDL, NJ
2016	Delivered science outreach presentations to years 10-12, King Solomon Academy school, London UK
2012-2013	Assisted in the Department of Geology open days at the University of Leicester | Responsible for welcoming members of the public to the department and also talking to prospective students about the Geology with Geophysics BSc programme

Technical Skills & tools

- Ocean / Sea ice models: Modular Ocean Model version 6 (MOM6) / Sea Ice Simulator version 2 (SIS2)
- Coupled climate models: Seamless system for Prediction and EArth system Research (SPEAR)
- Statistical techniques: Gaussian processes, neural networks, principal component analysis, relevance vector machines, complex networks, data assimilation
- Programming languages: Python (incl. PyTorch and Tensorflow), Matlab, Fortran-90